

2022 Ph H1 Q1

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

Question Summary

A ball is thrown vertically upwards and falls back to its starting position.

An acceleration-time graph is shown.

Which **velocity-time graph** represents the same motion?

 **Final Answer:**

A

Working

- Acceleration due to gravity remains constant and negative during both the upward and downward motion.
 - The velocity-time graph should be a straight line with **negative gradient**, crossing zero (at the highest point).
 - Only **graph A** matches this behaviour.
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Quick Tips

- In vertical motion, velocity changes sign (positive up, negative down).
 - Constant acceleration gives a **straight line** on a $v-t$ graph.
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